

TU Dortmund University — Department of Statistics

Case Studies – Summer Semester 2026

Case Studies – Project 3

Forecasting the COVID Series

Comparisons and Conclusions

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July 14, 2026

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1 Introduction

The COVID-19 pandemic created a large daily dataset for Germany. In this report, we use the daily number of newly reported COVID-19 cases from the Robert Koch-Institut (RKI). This series is useful for forecasting because it has clear time-series patterns. For example, case numbers depend strongly on past case numbers, and there is also a weekly reporting pattern. The series also changes over time because of testing behaviour, policy changes, variants, temperature, and the age structure of reported cases. Short-term forecasts can be useful for health planning, such as hospital capacity and resource allocation (Petropoulos and Makridakis, 2020).

This report is the third part of the case study on forecasting daily COVID-19 cases in Germany. In Project 1, we mainly used linear autoregressive models. These models used past case values and some extra variables, such as age-group variables, temperature, and a Monday dummy. In Project 2, we added machine learning models, mainly decision trees and random forests.

In this project, we compare the models from the earlier projects on the same test period. The models include AR models, ARX models, decision trees, and random forests. We also compare two forecasting schemes. The first one is fixed-origin forecasting, where the model is estimated once. The second one is rolling-window forecasting, where the model is updated before each forecast. We also use the Diebold-Mariano test to check whether the forecast differences are statistically meaningful.

First, we split the log-transformed case series into training and test data. The first 80 percent of the observations are used for training. The last 20 percent are used for testing. The lag order is chosen using the ACF, PACF, and BIC.

After that, we produce one-step-ahead forecasts for the test period. In the fixed-origin scheme, the model is estimated once using the training data, and the model parameters stay fixed during the test period. However, each one-step-ahead forecast still uses the actual past observations available at that date. In the rolling-window scheme, the model is re-estimated before each forecast using the most recent observations.

For the ARX models, we use age-group shares, temperature, and the Monday dummy. These covariates are first added separately. We also include one ARX model where all covariates are used together. For the machine learning models, we use one version with only lagged case values and one version with lagged case values plus covariates.

In total, this report compares 15 forecast series. All forecasts are evaluated on the same test period. Forecast accuracy is measured using RMSFE on the original case-count scale. The models are ranked by their RMSFE values. The Diebold-Mariano test is then used to compare the forecast errors. Since many model pairs are compared, Bonferroni-adjusted p -values are also reported.

The rest of the report is organised as follows. Section 2 describes the data and covari-

ates. Section 3 explains the methods. Section 4 presents the results. Section 5 summarises the main findings and limitations.

2 Problem and Data Description

2.1 Objectives

The aim of this project is to compare different one-step-ahead forecasts for daily COVID-19 cases in Germany. We use both linear models and machine learning models. The models mainly use past case values. Some models also use extra variables such as age-group shares, temperature, and the Monday dummy.

The first question is whether the extra variables improve the forecasts. For the ARX models, the covariates are first added separately. We also estimate one ARX model where all covariates are included together.

The second question is whether updating the model improves forecast accuracy. In the fixed-origin scheme, the model is estimated once using the training data, and the model parameters stay fixed during the test period. Each forecast still uses the actual past values available at that date. In the rolling-window scheme, the model is estimated again before each forecast using the most recent observations.

The third question is whether machine learning models perform better than the linear models. For this, we compare decision tree and random forest forecasts with the AR and ARX forecasts. All models are compared on the same test period using RMSFE.

2.2 Data source and collection

The case data come from the public COVID-19 dataset of the Robert Koch-Institut (RKI) (Robert Koch-Institut, 2026). The dataset contains reported COVID-19 cases in Germany. These cases are reported through the German public health system.

In the raw data, each row refers to reported cases with the same district, age group, sex, reporting date, and reference date. The raw dataset used in this project has 7,758,490 rows and 12 columns

For this project, the most important date variable is the reporting date, called *Melde-datum*. The data are administrative reporting data, not survey data. This means that the numbers show reported cases, not necessarily all true infections. The series can therefore be affected by reporting delays, weekday effects, and changes in testing behaviour.

To create the response variable, we aggregate `AnzahlFall` by reporting date. We then create a complete daily date series from April 1, 2020 to March 31, 2026. If a date has no observation after aggregation, the value is set to zero. This gives a daily case series with 2,191 observations.

The daily case numbers change a lot over time. Some days have only a few reported cases, while pandemic waves have much higher values. For this reason, we use the log transformation

$$y_t = \log(1 + \text{Cases}_t). \quad (1)$$

The value 1 is added so that days with zero reported cases can also be used. All models are estimated using this log-transformed series. The ACF and PACF plots are also based on the log series.

After forecasting, the predicted log values are transformed back to the original case-count scale:

$$\widehat{\text{Cases}}_t = \exp(\hat{y}_t) - 1. \quad (2)$$

This makes the RMSFE easier to interpret, because it is reported in daily case numbers instead of log points.

2.3 Covariates

Besides past case values, we use three types of covariates: age-group shares, temperature, and a Monday dummy. These variables are used in the ARX models and in the machine learning models.

First, we use the age-group information from the RKI data. The reported cases are divided into six age groups: A00-A04, A05-A14, A15-A34, A35-A59, A60-A79, and A80+ (Robert Koch-Institut, 2026). For each day, we calculate the share of cases in each age group.

Since the six age shares add up to one, we drop the youngest group, A00-A04, and use it as the reference group. This leaves five age-share variables. In the linear ARX models, these variables are lagged by one day, so that no information from the forecast day is used.

In the machine learning models, we create ten lags from the age-share variables. Since the age shares are already shifted by one day, these predictors refer to age shares from 2 to 11 days before the forecast date. This gives 50 age-share predictors.

Second, we use daily mean temperature for Berlin from the Open-Meteo historical weather archive (Open-Meteo, 2024). Temperature is included because it may be related to seasonal patterns in cases and reporting. Small gaps in the temperature series are filled before the models are estimated. In the ARX models, temperature is lagged by one day. In the machine learning models, ten temperature lags are used.

Third, we use a Monday dummy. This variable is equal to one on Mondays and zero otherwise. It is included because COVID-19 reporting has a clear weekly pattern. The

Monday dummy is not lagged, because the day of the week is already known in advance.

In the linear ARX models, age shares, temperature, and the Monday dummy are first included separately. We also estimate one ARX model with all covariates together. In the machine learning models with covariates, the full predictor set contains ten case lags, 50 age-share predictors, ten temperature lags, and the Monday dummy. This gives 71 predictors in total.

Table 1 gives a short summary of the constructed dataset and covariates.

Table 1: Summary of the constructed daily dataset

Characteristic	Value
Sample period	2020-04-01 to 2026-03-31
Number of daily observations	2,191
Response variable	$y_t = \log(1 + \text{Cases}_t)$
Age-share variables	5; reference group: A00–A04
Age shares in ARX models	Lagged by 1 day
Temperature in ARX models	Daily mean for Berlin, lagged by 1 day
Age shares in ML models	50 predictors, effective lags 2–11
Temperature in ML models	10 predictors, lags 1–10
Monday dummy	Equal to 1 on Mondays, not lagged
ML predictors with covariates	71 in total
Maximum case-series lag n	10
Train/test split	80% / 20% (chronological)

2.4 Data quality

There are some data quality issues in the data. First, the data are based on reported cases from the public health system and not on a random survey (Robert Koch-Institut, 2026). This means that the reported case numbers do not show the true number of infections perfectly. They depend on testing behaviour, reporting rules, reporting delays, and the capacity of the health authorities. These factors changed during the pandemic.

Second, the data have a clear weekly reporting pattern. Fewer cases are usually reported during weekends, while more cases may appear at the beginning of the next week. For this reason, the Monday dummy is included as a covariate. The maximum lag length is fixed at $n = 10$, which is enough to include a full weekly cycle.

Third, the sample covers different phases of the pandemic. These include different variants, vaccination, changes in testing behaviour, and the later period with lower reported case numbers. Because of this, the relationship between past cases and future cases may change over time. This is one reason why rolling-window estimation is useful.

There are also some issues with the age-group variables. Cases with unknown age group are not used when calculating the daily age shares. If age shares cannot be calculated for a day, the missing shares are replaced with zero in the code. This keeps the daily dataset complete, but the age-share variables should still be interpreted carefully.

Temperature values are also checked before the models are estimated. If temperature values are missing after merging with the case data, they are completed using nearby available values. Lag construction also creates missing values at the beginning of the sample, because earlier observations are not available. These rows are removed before the models are estimated.

Finally, all forecasts are compared on matching test-period dates. Before calculating RMSFE, observations with missing actual values or missing forecasts are removed. For the Diebold-Mariano tests, each pair of models is compared only on dates where the actual value and both forecasts are available. This makes sure that the forecast errors are compared fairly.

3 Methods

All analysis is done in Python. The data are prepared using `pandas` and `numpy` (McKinney, 2010; Harris et al., 2020). The AR models, ARX models, and the ACF and PACF plots are produced using `statsmodels` (Seabold and Perktold, 2010).

The decision tree and random forest models are estimated using `scikit-learn` (Pedregosa et al., 2011). The same package is also used for time-series cross-validation and random hyperparameter search in the machine learning part.

The Diebold-Mariano test is calculated using `scipy.stats` (Virtanen et al., 2020). The figures are created using `matplotlib` and `seaborn` (Hunter, 2007; Waskom, 2021).

3.1 General forecasting setup

The main variable in this project is the log-transformed daily case series, denoted by y_t . All models use past values of this series. Some models also use extra variables, such as age-group shares, temperature, and the Monday dummy.

The general forecasting model can be written as

$$y_t = f(y_{t-1}, \dots, y_{t-n}, Z_t) + \varepsilon_t, \quad (3)$$

where Z_t contains the covariates used in the model. The function $f(\cdot)$ is linear for the AR and ARX models. For the decision tree and random forest models, it is non-linear.

At each forecast date, the model makes a one-step-ahead forecast. This means that the model uses past case values and the available covariates to predict the next value:

$$\hat{y}_{t+1} = \hat{f}_t(y_t, \dots, y_{t-n+1}, Z_{t+1}). \quad (4)$$

The maximum number of case lags is fixed at $n = 10$. The linear models use either 7 or 10 case lags. The machine learning models use ten lagged case values. The covariates are used differently in the linear and machine learning models, as explained in the next subsections.

The data are split in chronological order. The first 80% of observations are used for training, and the last 20% are used for testing. This gives 1,752 training observations from April 1, 2020 to January 16, 2025, and 439 test observations from January 17, 2025 to March 31, 2026.

All models are evaluated on the same test period. Before comparing accuracy, the forecasts are matched by date.

3.2 Autoregressive and ARX models

We first use autoregressive models. An autoregressive model uses past values of the same series to predict the current value (Hamilton, 1994). An $AR(p)$ model is written as

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t, \quad (5)$$

where p is the number of lags, c is the constant, ϕ_i is the coefficient for lag i , and ε_t is the error term.

The lag order is chosen in two steps. First, we look at the ACF and PACF plots to see the main pattern in the data. Since the case series has a clear weekly pattern, $AR(7)$ is used as a benchmark model.

Second, we use the Bayesian information criterion (BIC) to compare AR models with lag orders from $p = 1$ to $p = 10$ (Schwarz, 1978). The maximum lag is limited to 10 because this is required in the project instructions. In the results, the lowest BIC is obtained for $AR(10)$.

We also estimate ARX models. ARX models extend the AR model by adding covariates. The model is written as

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \beta' Z_t + \varepsilon_t, \quad (6)$$

where Z_t contains the covariates used in the model.

All ARX models use ten case lags. We estimate four ARX specifications. The first adds the five age-share variables lagged by one day. The second adds daily mean temperature lagged by one day. The third adds the Monday dummy. The Monday dummy is not lagged because the weekday of the forecast date is already known. The fourth ARX

model includes all covariates together.

This allows us to compare the separate effect of each covariate group and also the combined effect of all covariates. The AR and ARX models are estimated using `statsmodels`.

3.3 Fixed-origin versus rolling-window estimation

We use two forecasting schemes in this project. The first one is the fixed-origin scheme. In this approach, the model is estimated once using the training sample. After that, the model parameters are kept fixed during the test period.

The forecasts are still one-step-ahead forecasts. For each test date, the model uses the actual past observations available up to that date. This means that new observations can enter the model as lagged values, but the model itself is not estimated again. In this report, fixed-origin means that the model parameters are fixed, but the lagged input values are updated with the actual observed data.

The second approach is the rolling-window scheme. In this approach, the model is estimated again before each forecast. Each time, the model uses the most recent 1,752 observations. This is the same size as the original training sample. After one forecast is made, the window moves forward by one day. The oldest observation is dropped, and the newest available observation is added.

The rolling-window method is used for the AR model, the ARX models, and the machine learning models. All rolling models use the same window length and forecast the same test dates. This makes the results comparable.

Rolling-window estimation takes more time because the models are fitted many times. However, it allows the model parameters to change when the case series changes over time. Comparing fixed-origin and rolling-window forecasts shows whether this model updating improves forecast accuracy.

3.4 Machine learning methods

In addition to the linear models, we use two machine learning methods. These are a decision tree and a random forest. Both methods are tree-based methods and follow the general idea used in Project 2.

The first method is decision tree regression. A decision tree splits the data into smaller groups using the predictor variables. The final prediction is the average value of the training observations in the selected group (Breiman et al., 1984).

The second method is random forest regression. A random forest uses many decision trees and averages their forecasts. This usually makes the prediction more stable than using only one tree (Breiman, 2001).

The main model settings are tuned using `RandomizedSearchCV`. For the decision tree, the tuning checks different values for tree depth and the minimum number of observations

in a leaf. For the random forest, it checks different numbers of trees, tree depths, and numbers of predictors used at each split.

The tuning uses three-fold time-series cross-validation. This is useful because it keeps the order of the data. Twenty random hyperparameter combinations are tested in each tuning step. The tuning is repeated every 14 forecast days. Between these tuning dates, the selected settings are kept, but the model is still fitted again each day using the latest rolling window.

Both machine learning methods use a rolling window with a maximum length of 1,752 observations. For each forecast date, only observations before that date are used to train the model.

For Part (e), the decision tree and random forest use only the ten lagged values of the log case series. For Part (f), the same ten case lags are used together with the covariates. The full covariate version includes 50 age-share predictors, ten temperature lags, and the Monday dummy. Together with the ten case lags, this gives 71 predictors.

These two feature sets allow us to check whether adding age-group information, temperature, and the Monday dummy improves the machine learning forecasts. All machine learning forecasts are compared on the same test period as the linear forecasts.

3.5 Forecast evaluation: RMSFE

To compare the forecasts, we use the root mean squared forecast error (RMSFE). The models are estimated using the log-transformed case series, but the forecast errors are calculated on the original case-count scale. For this reason, the predicted log values are first transformed back into daily case numbers:

$$\widehat{\text{Cases}}_t = \exp(\hat{y}_t) - 1. \quad (7)$$

The RMSFE is calculated as

$$\text{RMSFE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(\text{Cases}_t - \widehat{\text{Cases}}_t \right)^2}, \quad (8)$$

where T is the number of test-period dates with both an actual value and a forecast. If a forecast is missing for a date, that date is not used for that model.

A lower RMSFE means that the model has smaller forecast errors and performs better. Because the errors are squared, large errors have more influence on the RMSFE than small errors.

We report RMSFE on the original case-count scale because it is easier to understand. The values can then be read as forecast errors in daily reported cases instead of log points.

3.6 The Diebold–Mariano test

To check whether forecast differences are statistically meaningful, we use the Diebold–Mariano test (Diebold and Mariano, 1995; Diebold, 2012). This test compares the forecast errors of two models. Since RMSFE is based on squared errors, we also use squared errors in the Diebold–Mariano test.

Let $\hat{A}_t^{(1)}$ and $\hat{A}_t^{(2)}$ be forecasts from two models for the same actual value A_t . The loss difference is

$$d_t = \left(A_t - \hat{A}_t^{(1)}\right)^2 - \left(A_t - \hat{A}_t^{(2)}\right)^2. \quad (9)$$

If the two models have similar forecast accuracy, the average value of d_t should be close to zero. A negative value means that the first model has lower squared error on average. A positive value means that the second model has lower squared error on average.

The Diebold–Mariano statistic is calculated as

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}, \quad \bar{d} = \frac{1}{T} \sum_{t=1}^T d_t, \quad (10)$$

where T is the number of dates where the actual value and both forecasts are available. Each model pair is therefore compared on the same dates.

All forecasts in this report are one-step-ahead forecasts, so the forecast horizon is $h = 1$. We also use the small-sample correction of Harvey, Leybourne and Newbold (Harvey et al., 1997):

$$\text{DM}^* = \text{DM} \sqrt{\frac{T + 1 - 2h + h(h - 1)/T}{T}}. \quad (11)$$

The corrected statistic is compared with a Student- t distribution with $T - 1$ degrees of freedom. In the DM heatmap later in the report, a negative value means that the row model has lower average squared error than the column model.

Since the final comparison includes 15 forecast models, there are 105 pairwise comparisons. We therefore report both raw p -values and Bonferroni-adjusted p -values. The Bonferroni correction makes the test stricter because many model pairs are compared.

4 Empirical Results

4.1 Autoregressive order selection

Figure 1 shows the ACF and PACF plots of the log case series for the first ten lags. The series depends strongly on recent past values. There is also a visible pattern around lag 7, especially in the PACF. This fits the weekly reporting cycle in the COVID-19 data.

Because of this, the AR(7) model is used as a weekly benchmark.

We also use the Bayesian information criterion (BIC) to choose the lag order more formally (Schwarz, 1978). The BIC is calculated for AR models with lag orders from $p = 1$ to $p = 10$. Figure 2 shows that the lowest BIC is obtained at $p = 10$. Therefore, within the allowed lag range, BIC selects the AR(10) model.

For this reason, we report both AR(7) and AR(10). The AR(7) model is included because it represents the weekly reporting pattern. The AR(10) model is the BIC-selected specification and is also used as the autoregressive part of the ARX models. The machine learning models also use ten lagged values of the case series.

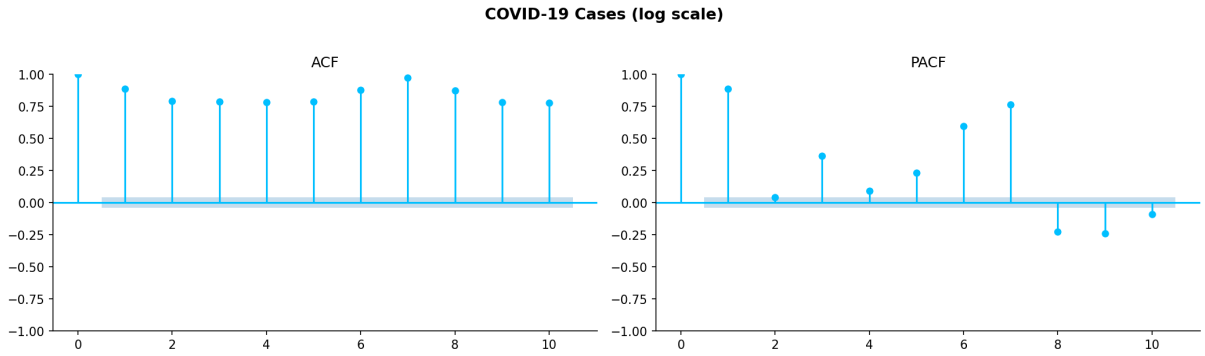


Figure 1: ACF and PACF plots of the log case series for the first ten lags. The plots show strong short-term dependence and a weekly pattern around lag 7.

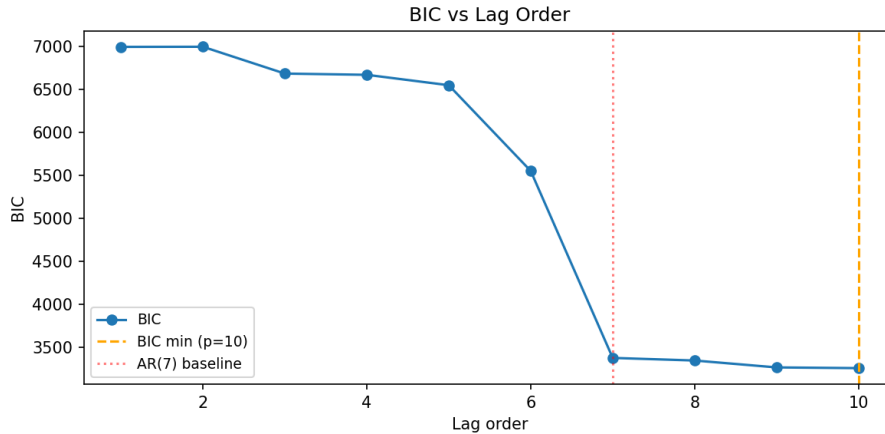


Figure 2: BIC values for AR models with lag orders from $p = 1$ to $p = 10$. The lowest BIC is obtained at $p = 10$, while $p = 7$ is included as the weekly benchmark.

4.2 Part (b): Fixed-origin autoregressive forecasts

Table 2 and Figure 3 show the fixed-origin forecasts for the AR(7) and AR(10) models. Both models are estimated once using the initial training sample. Their estimated coefficients are then kept fixed during the test period.

However, each forecast is still made one step ahead using the actual past observations available at that date. This means that new case observations enter the model as lagged values, but the model parameters are not estimated again.

Table 2: RMSFE of fixed-origin autoregressive forecasts

Model	RMSFE
AR(7), fixed	223.73
AR(10), fixed	199.73

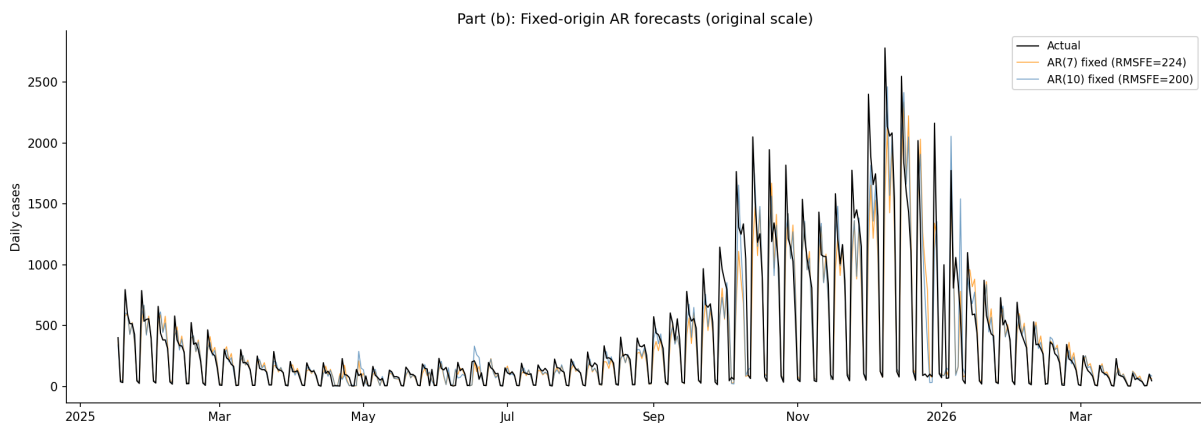


Figure 3: Actual cases and fixed-origin one-step-ahead AR forecasts during the test period.

The AR(10) model performs better than the AR(7) model. Its RMSFE is 199.73, compared with 223.73 for AR(7). This suggests that the three additional lags contain useful information beyond the weekly AR(7) structure.

Figure 3 shows that both models follow the general movement of the actual case series. However, AR(10) is closer to the observed values in several parts of the test period. This is consistent with its lower RMSFE. Since the model parameters are fixed in both cases, the difference comes from the lag structure rather than from model updating.

4.3 Part (c): Fixed-origin ARX forecasts

Table 3 and Figure 4 show the fixed-origin results for the ARX(10) models. Each model extends the AR(10) specification by adding age shares, temperature, the Monday dummy, or all covariates together.

The models are estimated once using the initial training sample, and their estimated parameters are then kept fixed. However, each forecast is produced one step ahead using the actual past observations available at that date.

Table 3: RMSFE of fixed-origin ARX(10) forecasts

Model	RMSFE
ARX(10) + Age shares	198.97
ARX(10) + Temperature	199.60
ARX(10) + Monday dummy	207.38
ARX(10) + All covariates	206.61

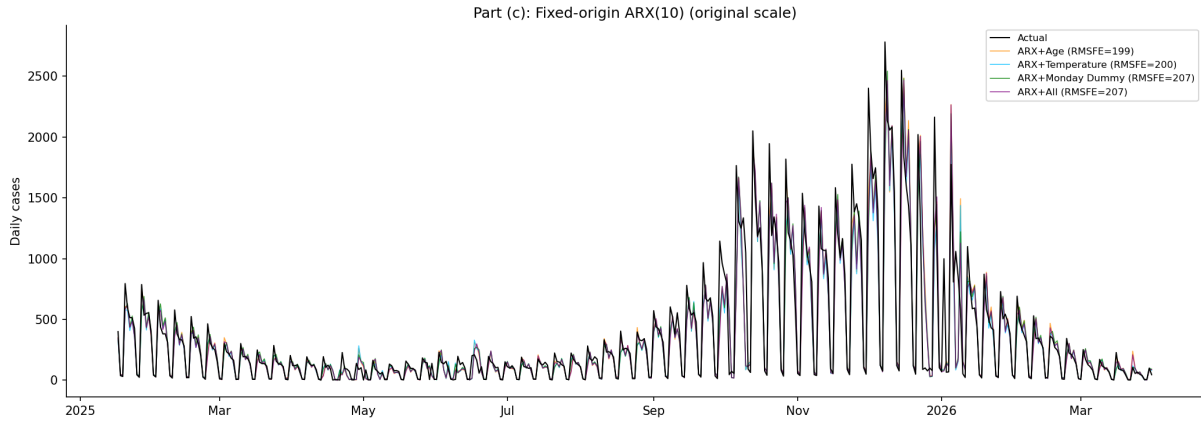


Figure 4: Actual cases and fixed-origin one-step-ahead ARX(10) forecasts during the test period.

The best fixed-origin ARX model is the model with age shares, with an RMSFE of 198.97. This is slightly lower than the RMSFE of 199.73 for the fixed-origin AR(10) model. However, the improvement is small.

The temperature model also performs very similarly to the plain AR(10) model, with an RMSFE of 199.60. This suggests that temperature adds little additional forecasting information in this specification.

The models containing the Monday dummy perform somewhat worse. The Monday-only model has an RMSFE of 207.38, while the model containing all covariates has an RMSFE of 206.61. Therefore, adding more covariates does not automatically improve forecast accuracy. Overall, the differences between the fixed-origin AR(10), age-share, and temperature models are quite small.

4.4 Part (d): Rolling-window linear forecasts

Table 4 and Figure 5 show the results for the rolling-window linear models. In this part, the AR(10) and ARX(10) models are re-estimated before each forecast. The rolling window has a fixed length of 1,752 observations, which is the same size as the initial training sample.

Table 4: RMSFE of rolling-window AR(10) and ARX(10) forecasts

Model	RMSFE
AR(10), rolling	202.13
ARX(10) + Age shares, rolling	201.78
ARX(10) + Temperature, rolling	202.18
ARX(10) + Monday dummy, rolling	198.33
ARX(10) + All covariates, rolling	197.16

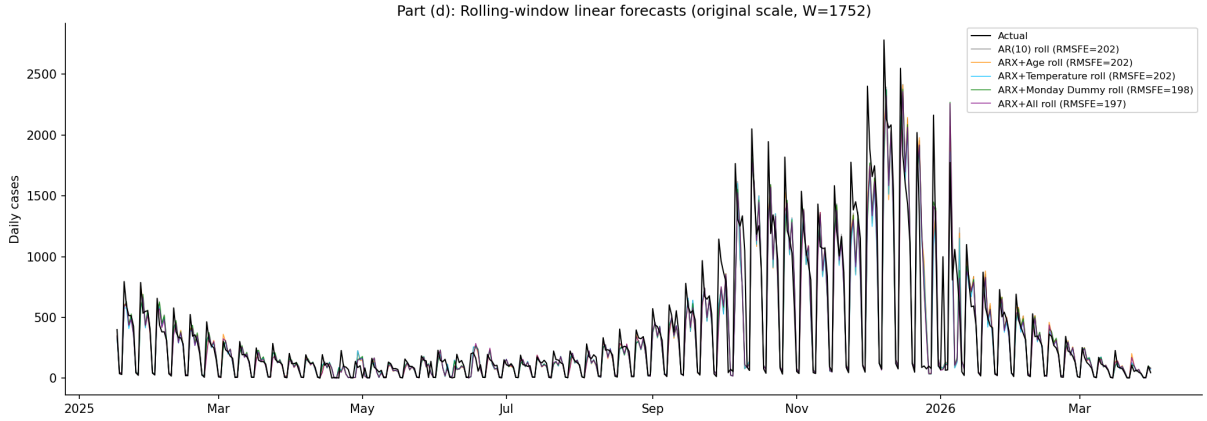


Figure 5: Actual cases and rolling-window AR(10) and ARX(10) forecasts during the test period.

The best rolling linear model is the ARX(10) model with all covariates. It has an RMSFE of 197.16. The model with the Monday dummy is second, with an RMSFE of 198.33.

The plain rolling AR(10) model has an RMSFE of 202.13. Adding age shares gives a small improvement to 201.78, while adding temperature gives a slightly higher RMSFE of 202.18. Therefore, the differences between these three rolling models are small.

The comparison with the fixed-origin results is mixed. The rolling-window method improves the Monday model from 207.38 to 198.33 and the model with all covariates from 206.61 to 197.16. However, the fixed-origin AR(10), age-share, and temperature models perform slightly better than their rolling versions.

These results show that updating the model does not improve every specification. Rolling-window estimation is most useful for the models containing the Monday dummy and the full covariate set. Overall, both the estimation scheme and the choice of covariates affect forecast performance, but the differences between the strongest linear models are quite small.

Figure 6 compares all linear models considered so far and ranks them by RMSFE.

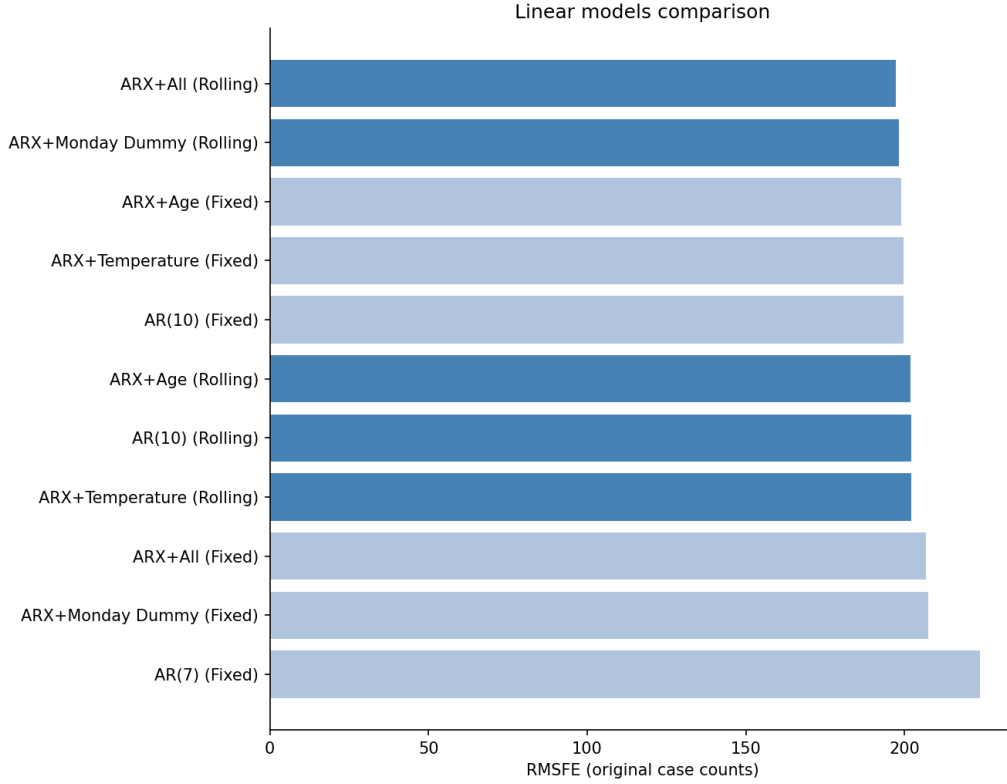


Figure 6: Comparison of fixed-origin and rolling-window linear models by RMSFE. Lower values indicate better forecast performance.

The full comparison shows that the rolling ARX model with all covariates is the best linear model. However, the rolling ARX model with the Monday dummy and the fixed-origin models with age shares, temperature, and ten case lags also have similar RMSFE values.

4.5 Parts (e) and (f): Machine learning forecasts

Table 5 shows the rolling-window results for the machine learning models. We estimate both the decision tree and random forest models in two ways. First, we use only the ten lagged values of the log case series. This corresponds to Part (e). Second, we add the full covariate set, including the lagged age-share variables, lagged temperature, and the Monday dummy. This corresponds to Part (f). The same rolling window of 1,752 observations is used as in the linear models.

Table 5: RMSFE of rolling-window machine learning forecasts

Model	Features	RMSFE
Decision tree	Lags only	194.51
Random forest	Lags only	189.59
Decision tree	Lags + covariates	220.07
Random forest	Lags + covariates	196.97

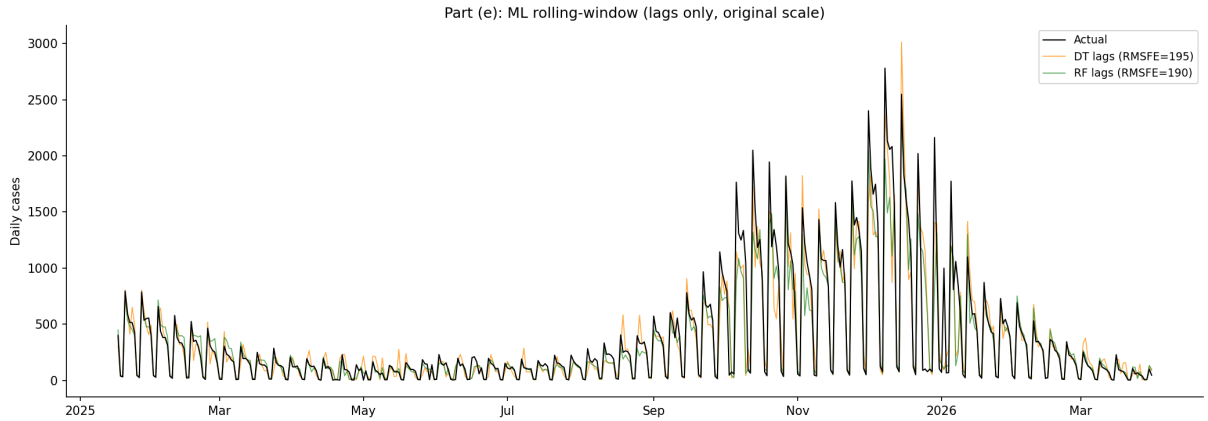


Figure 7: Actual cases and rolling-window machine learning forecasts using lagged case values only.

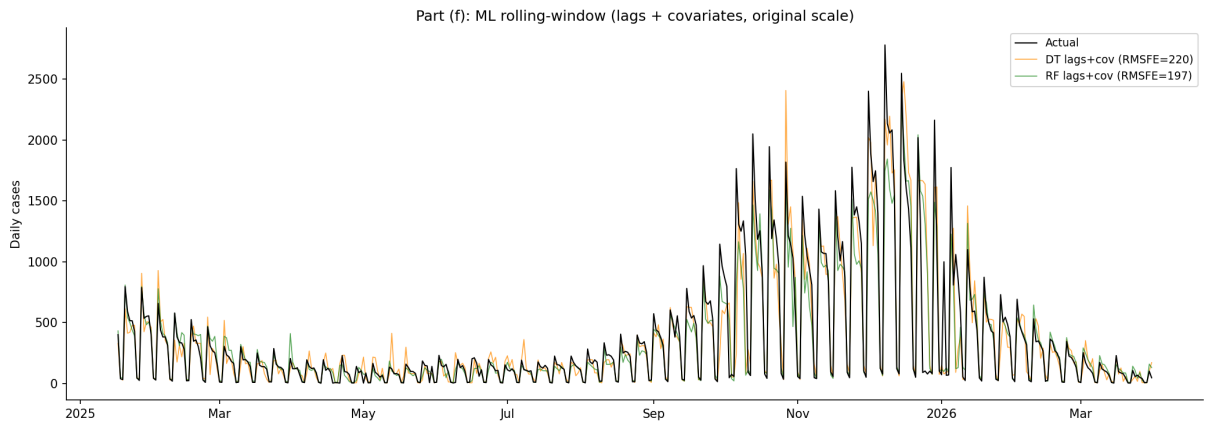


Figure 8: Actual cases and rolling-window machine learning forecasts using lagged case values and covariates.

The random forest performs better than the decision tree in both specifications. With lagged case values only, the decision tree has an RMSFE of 194.51, while the random forest has a lower RMSFE of 189.59. When the covariates are added, the decision tree has an RMSFE of 220.07 and the random forest has an RMSFE of 196.97.

Adding the covariates does not improve either machine learning model. For the decision tree, the RMSFE increases from 194.51 to 220.07. Figure 8 also shows that the

decision tree with covariates is less stable than the lags-only version. This suggests that the additional predictors do not help the single tree in this setup.

The random forest with covariates performs better than the decision tree with covariates, but it is still worse than the random forest using only lagged case values. Its RMSFE increases from 189.59 to 196.97 after adding the covariates.

Overall, the results suggest that the past case values already contain most of the useful information for the machine learning forecasts. The best machine learning model is the random forest with lags only. Its RMSFE of 189.59 is also the lowest value among all models considered so far.

4.6 Feature importance in the machine learning models

To examine which variables are most important in the machine learning models, Figure 9 shows the feature importance values for the decision tree and random forest. For this interpretation, both models are fitted once on the full training sample using the lagged case values and covariates. The selected hyperparameters from the tuning procedure are used. Figure 10 shows the first three levels of the fitted decision tree.

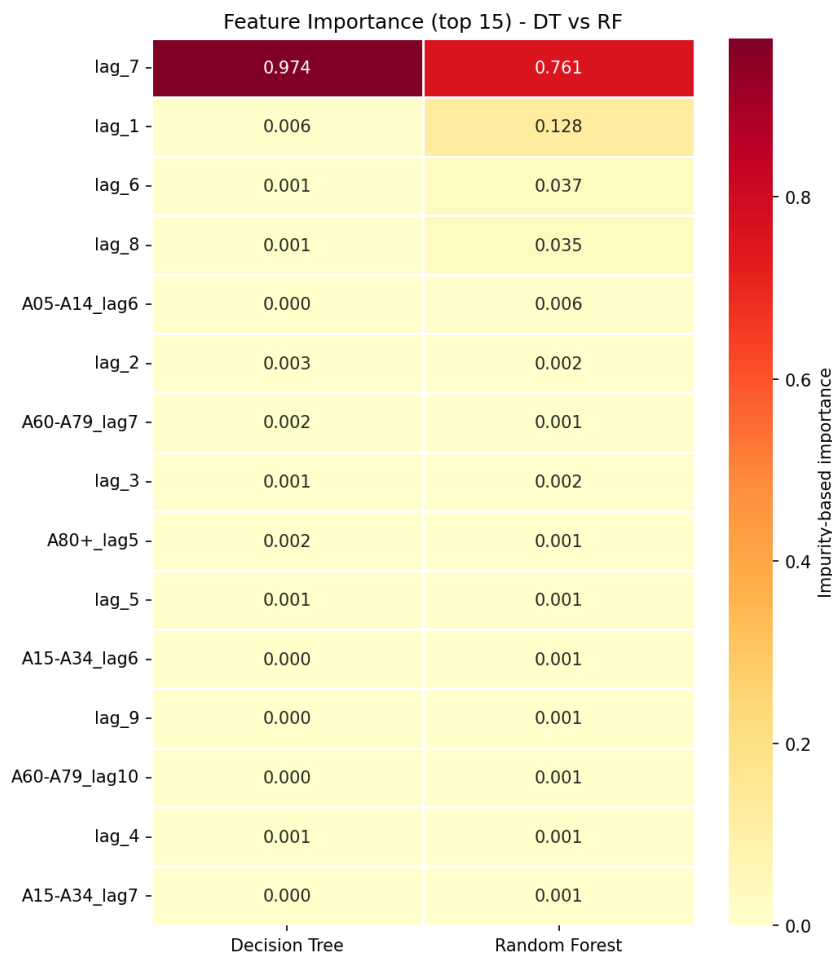


Figure 9: Feature importance for the decision tree and random forest models using lagged case values and covariates.

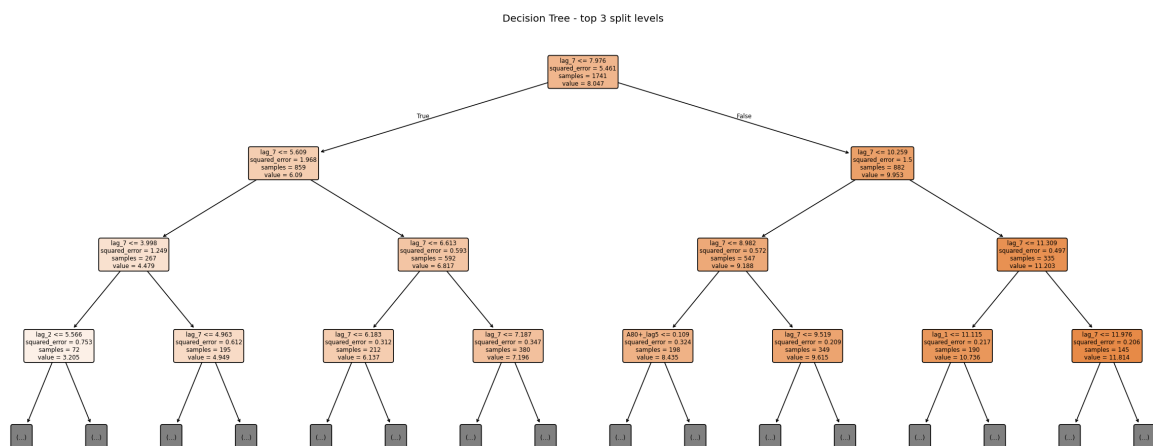


Figure 10: First three split levels of the fitted decision tree. The tree is shown only partly for readability.

The results show that lag 7 is the most important predictor in both models. This means that the reported case value from exactly one week earlier is the main variable

used for prediction. In the decision tree, lag 7 accounts for about 97.39% of the total feature importance. In the random forest, its importance is about 76.08%.

In the random forest, lag 1 is the second most important variable, with an importance of about 12.78%. Lag 6 and lag 8 also have some importance, with values of about 3.67% and 3.49%. The remaining case lags and covariates have much smaller importance values.

The decision tree relies much more strongly on lag 7 than the random forest. The random forest spreads some importance across other lagged case values. However, in both models, the age-share variables, temperature variables, and Monday dummy contribute only a small amount.

This result is consistent with the ACF and PACF plots in Figure 1. The weekly reporting pattern is the strongest source of predictability in the case series. Once lag 7 and the nearby case lags are included, the extra covariates provide only limited additional information.

The importance values describe how strongly the variables are used for splitting the training data. They should not be interpreted as causal effects.

4.7 Part (g): Overall RMSFE ranking

Table 6 and Figure 11 show the full RMSFE ranking for all fifteen forecast models. All models are evaluated over the same 439-day test period, so their RMSFE values can be compared directly.

Table 6: Full RMSFE ranking of all models

Rank	Model	RMSFE
1	Random forest, lags only, rolling	189.59
2	Decision tree, lags only, rolling	194.51
3	Random forest, lags + covariates, rolling	196.97
4	ARX(10) + All covariates, rolling	197.16
5	ARX(10) + Monday dummy, rolling	198.33
6	ARX(10) + Age shares, fixed-origin	198.97
7	ARX(10) + Temperature, fixed-origin	199.60
8	AR(10), fixed-origin	199.73
9	ARX(10) + Age shares, rolling	201.78
10	AR(10), rolling	202.13
11	ARX(10) + Temperature, rolling	202.18
12	ARX(10) + All covariates, fixed-origin	206.61
13	ARX(10) + Monday dummy, fixed-origin	207.38
14	Decision tree, lags + covariates, rolling	220.07
15	AR(7), fixed-origin	223.73

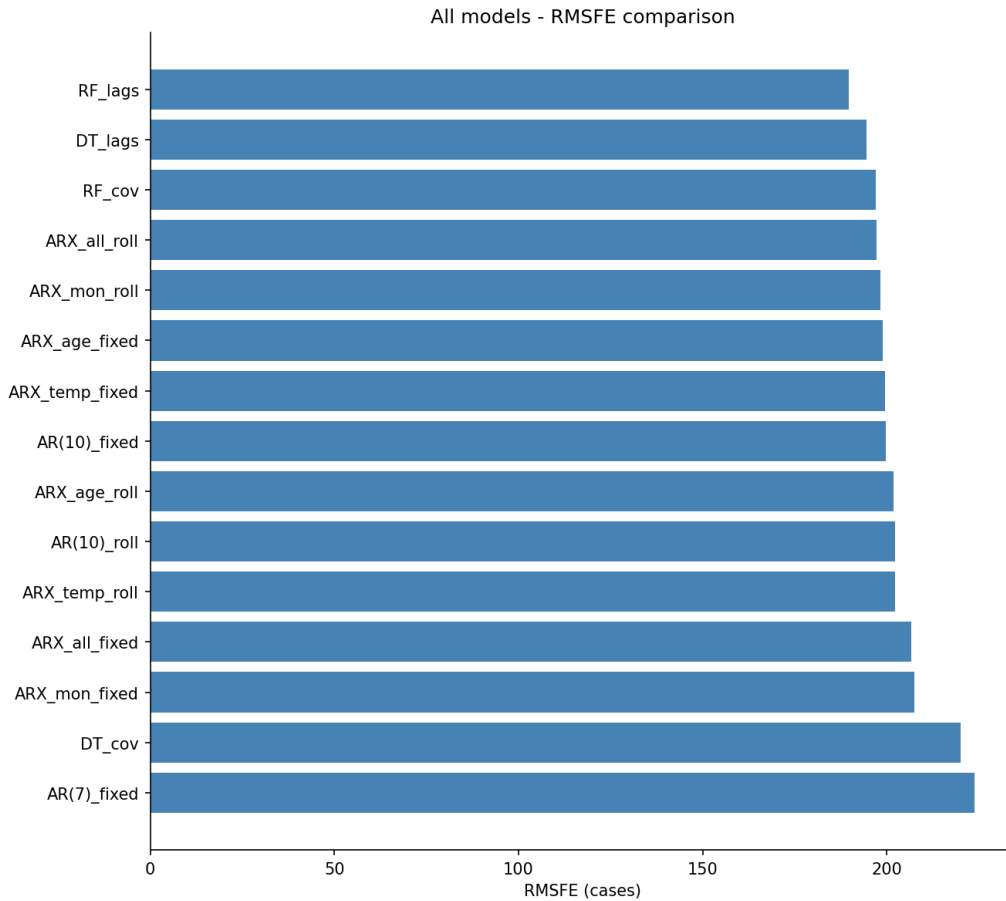


Figure 11: RMSFE comparison of all fifteen forecast models. Lower values indicate better forecast performance.

The best model is the rolling random forest using only the ten lagged case values. It has an RMSFE of 189.59. The decision tree with lags only is ranked second, with an RMSFE of 194.51. The random forest with covariates is third, with an RMSFE of 196.97.

Adding covariates does not improve the machine learning models. For the random forest, the RMSFE increases from 189.59 to 196.97. For the decision tree, the RMSFE increases from 194.51 to 220.07. This shows that the additional predictors do not automatically improve forecast accuracy.

The best linear model is the rolling ARX(10) model with all covariates, which has an RMSFE of 197.16. The rolling model with the Monday dummy and the fixed-origin age-share, temperature, and AR(10) models also perform similarly, with RMSFE values between about 198 and 200.

The ranking does not show a general advantage of rolling-window estimation over fixed-origin estimation. Some rolling models perform better than their fixed-origin versions, especially the models containing the Monday dummy or all covariates. However, the fixed-origin AR(10), age-share, and temperature models perform slightly better than their corresponding rolling versions.

Overall, the random forest with lags only gives the lowest RMSFE. However, several

other models are not far behind, especially the lags-only decision tree, the random forest with covariates, and the strongest rolling ARX models. The weakest models are the decision tree with covariates and the fixed-origin AR(7) benchmark. The Diebold–Mariano tests in the next subsection examine whether these differences are statistically significant.

Figure 12 compares the actual case series with five selected forecasts. The selected models include fixed-origin and rolling-window linear forecasts together with machine learning forecasts.

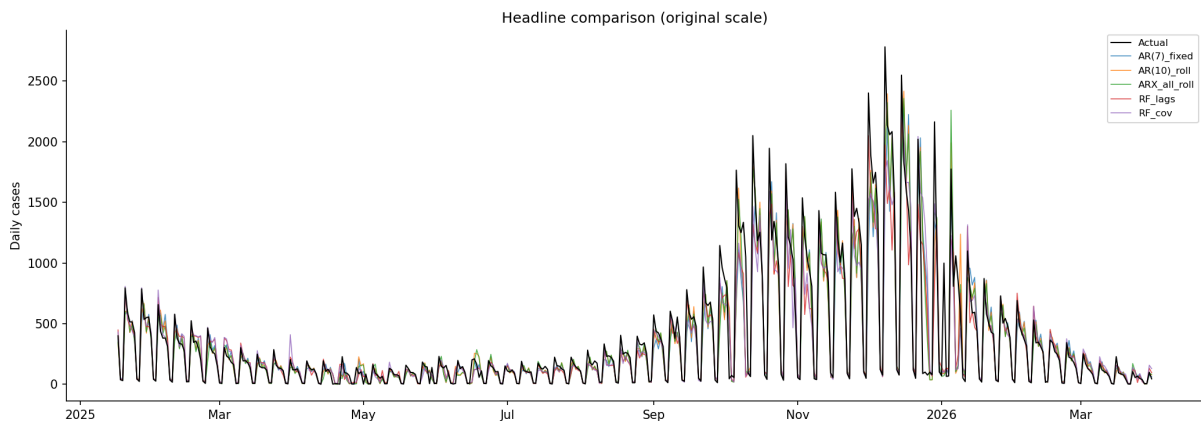


Figure 12: Actual cases and five selected forecasts during the test period.

Figure 12 shows that most models follow the main weekly movement of the case series. The largest forecast errors occur during periods with higher case numbers. The selected forecasts are generally close to each other, but the ranking in Table 6 shows that the random forest with lagged case values gives the lowest overall RMSFE.

4.8 Diebold–Mariano test results

Next, we use the Diebold–Mariano test to check whether the differences in forecast accuracy are statistically significant. Since there are 15 forecast models, this gives 105 unique pairwise comparisons. Figures 13, 14, and 15 show the DM statistics, the raw p -values, and the Bonferroni-adjusted p -values.

Using the raw p -values, 15 out of 105 comparisons are significant at the 5% level. After applying the Bonferroni correction, only one comparison remains significant. This shows that most differences in the RMSFE ranking are not strong enough to remain statistically significant after correcting for the large number of tests.

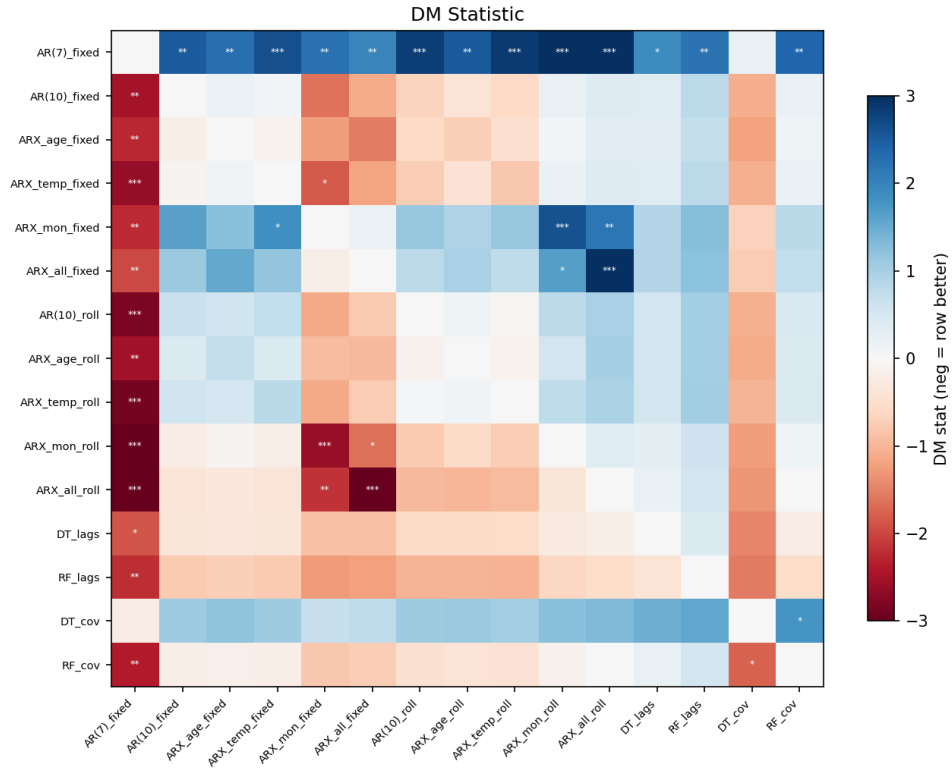


Figure 13: Diebold–Mariano statistics for all 15 models. Negative values mean that the row model has a lower average squared forecast error than the column model.

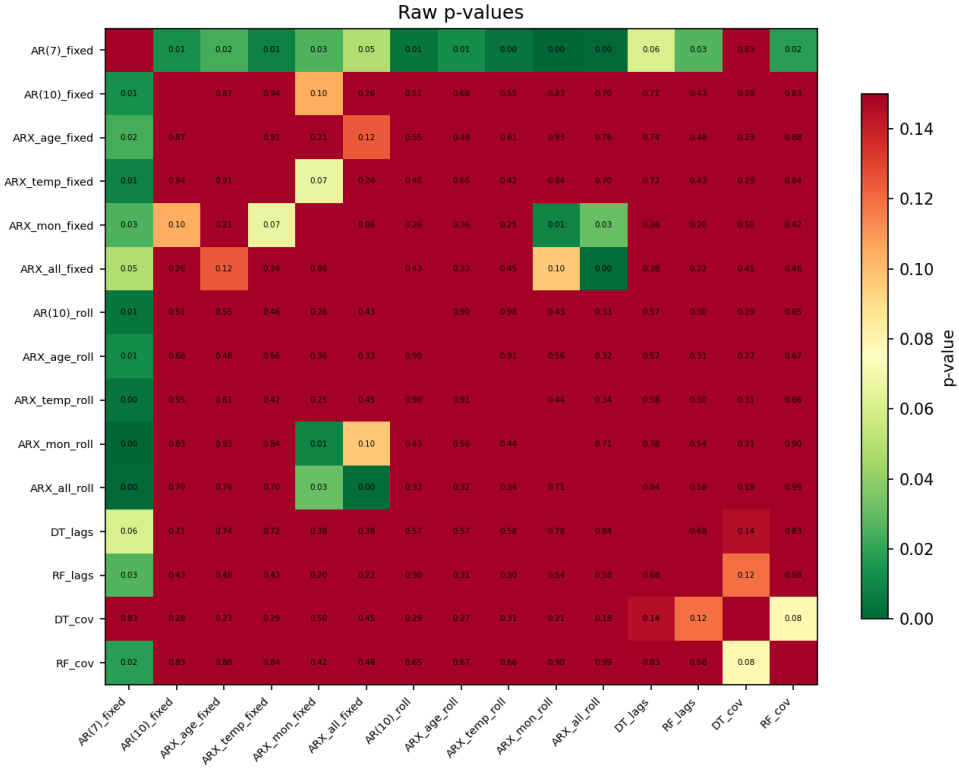


Figure 14: Raw p -values from the Diebold–Mariano tests for all 15 models.

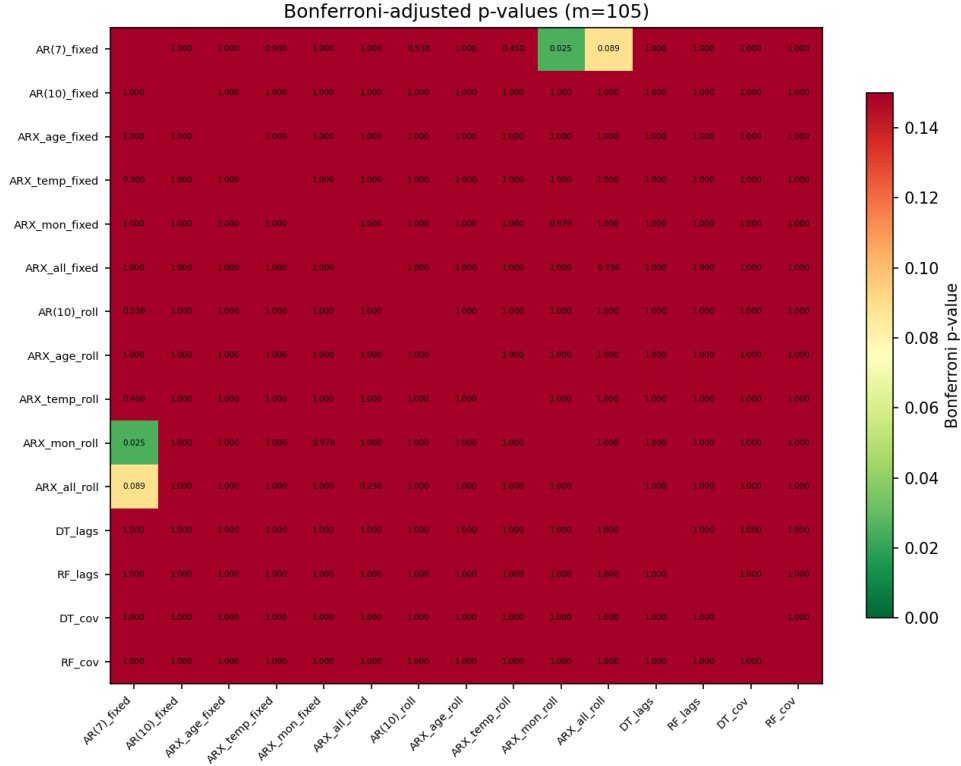


Figure 15: Bonferroni-adjusted p -values from the Diebold–Mariano tests for all 15 models.

The only comparison that remains significant after the Bonferroni correction is between the rolling ARX(10) model with the Monday dummy and the fixed-origin AR(7) model. The test indicates that the rolling ARX model with the Monday dummy has the lower average squared forecast error.

The random forest with lags only has the lowest RMSFE in the overall ranking. However, its advantage over the other stronger models is not statistically significant after the Bonferroni correction. Therefore, the RMSFE ranking shows which model performs best numerically, but the DM test does not provide strong evidence that it is clearly better than most competing models.

The results also do not show that rolling-window models generally outperform fixed-origin models. Some rolling models have lower RMSFE values, while some fixed-origin models perform slightly better than their rolling versions. Most of these differences are not statistically significant.

Overall, the DM test gives a more cautious conclusion than the RMSFE ranking. The models have different numerical forecast errors, but many of the stronger models perform similarly once multiple testing is taken into account.

4.9 Error diagnostics over time

Finally, we examine how the forecast errors change during the test period. Figure 16 shows the cumulative squared forecast error for five selected models. Figure 17 shows the

rolling 90-day RMSFE for the same models.

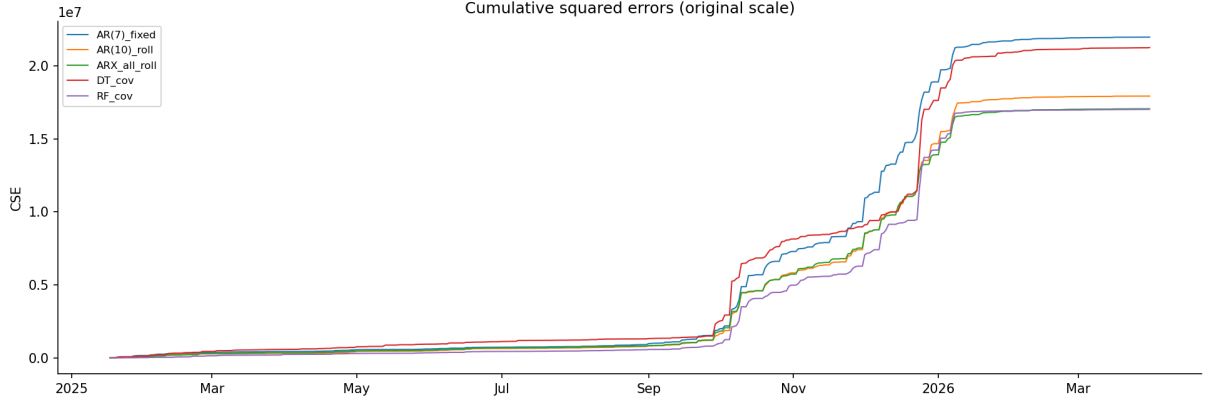


Figure 16: Cumulative squared forecast errors during the test period for five selected models.

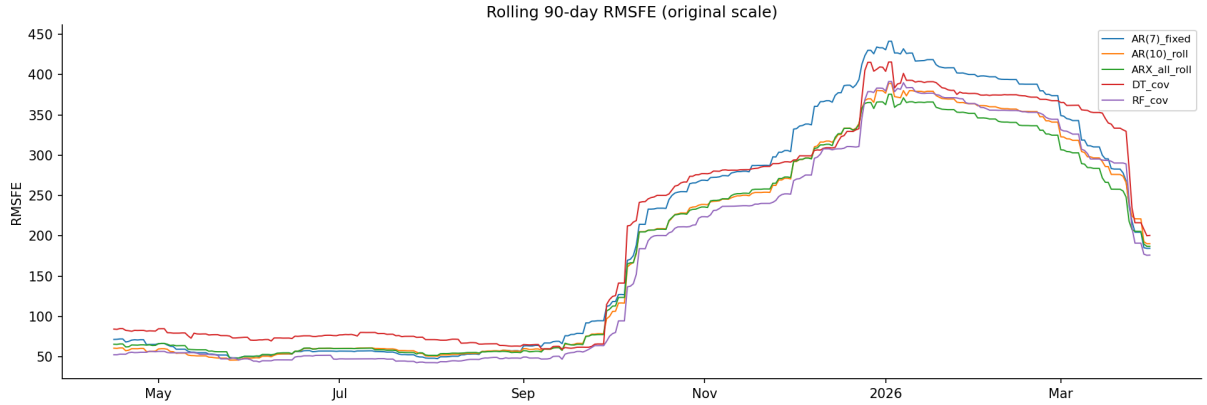


Figure 17: Rolling 90-day RMSFE during the test period for five selected models.

The cumulative squared-error plot shows that forecast errors increase slowly during the first part of the test period. The largest increases occur from around October 2025 to January 2026, when the reported case numbers are higher and change more strongly.

Among the five selected models, the fixed-origin AR(7) model has the highest cumulative squared error by the end of the test period. The decision tree with covariates also produces a high cumulative error, especially during the period with higher case numbers. The rolling ARX model with all covariates and the random forest with covariates have the lowest cumulative errors among the selected models. The rolling AR(10) model lies between these groups.

The rolling 90-day RMSFE plot shows a similar pattern. Forecast errors are relatively low during the first part of the test period, but they rise strongly from autumn 2025 and reach their highest values around the beginning of 2026. After this period, the rolling RMSFE decreases again as the reported case numbers become lower.

Overall, the figures show that forecast performance changes over time. Most of the total forecast error is created during the period with high and rapidly changing case

numbers. The differences between the models are smaller during periods with lower and more stable case numbers.

5 Conclusion

This project extends the forecasting work from Project 1 and Project 2. The aim was to compare linear and machine learning models for forecasting daily reported COVID-19 cases in Germany using RKI data (Robert Koch-Institut, 2026). The daily case series was log-transformed, and all models were evaluated over the same 439-day test period. We used AR and ARX models with up to ten case lags. BIC selected ten lags, while AR(7) was also included as a weekly benchmark (Schwarz, 1978). Decision trees and random forests were used as the machine learning methods (Breiman et al., 1984; Breiman, 2001). Forecast accuracy was measured using RMSFE on the original case-count scale, and the Diebold–Mariano test was used to compare forecast errors (Diebold and Mariano, 1995; Diebold, 2012; Harvey et al., 1997).

The overall results show that no single estimation scheme is always better. Rolling-window estimation improves some models, especially the ARX models containing the Monday dummy or all covariates. The combined ARX model improves from an RMSFE of 206.61 in the fixed-origin scheme to 197.16 in the rolling-window scheme. The Monday model improves from 207.38 to 198.33.

However, rolling estimation does not improve every model. The fixed-origin AR(10) model has an RMSFE of 199.73, compared with 202.13 for the rolling AR(10) model. The fixed-origin age-share and temperature models also perform slightly better than their rolling versions. Therefore, the results do not show a general advantage of rolling-window estimation. The effect of updating depends on the model and the variables included.

The covariates give mixed results. Among the fixed-origin models, adding age shares gives a small improvement over the plain AR(10) model. The temperature model performs almost the same as AR(10), while the Monday dummy and the combined covariate model perform worse. Among the rolling models, the combined ARX model gives the best linear forecast, followed by the model with the Monday dummy. However, the differences between the strongest linear models are small.

The best model overall is the rolling random forest using only the ten lagged case values. It has an RMSFE of 189.59. The decision tree with lags only is second, with an RMSFE of 194.51. The random forest with covariates is third, with an RMSFE of 196.97. This shows that the machine learning models can perform well, but adding the covariates does not improve forecast accuracy.

The decision tree results are mixed. The decision tree with lags only performs well and is ranked second overall. However, after adding the covariates, its RMSFE increases to 220.07. This makes it one of the weakest models in the comparison. The forecast plot

also shows that this model is less stable during the period with higher case numbers.

The feature importance results support the importance of the weekly reporting pattern. Lag 7 is the most important variable in both machine learning models. It accounts for about 97.39% of the importance in the decision tree and 76.08% in the random forest. Lag 1 is also important in the random forest, with an importance of about 12.78%. Lag 6 and lag 8 also have some importance, but the remaining lags and covariates have much smaller importance values. This suggests that the case value from one week earlier contains a large part of the useful forecasting information.

The Diebold–Mariano results give a more cautious conclusion than the RMSFE ranking. With 15 models, there are 105 pairwise comparisons. Fifteen comparisons are significant using the raw p -values, but only one remains significant after the Bonferroni correction. The only comparison that remains significant is between the rolling ARX model with the Monday dummy and the fixed-origin AR(7) model. Therefore, although the random forest with lags only has the lowest RMSFE, its advantage over most of the stronger competing models is not statistically significant after correcting for multiple testing.

The error diagnostics also show that forecast performance changes over time. Errors are relatively small during periods with low and stable case numbers. They increase strongly from autumn 2025 to the beginning of 2026, when reported cases are higher and change more quickly. A large part of the total forecast error is created during this period.

There are several limitations and possible directions for further work. First, the data cover a long period from 2020 to 2026, during which testing behaviour, reporting rules, vaccination, and virus variants changed. One model may not capture all of these changes equally well. Second, the covariates used in this project are limited. Future work could include other variables, such as mobility, vaccination, wastewater, or variant data. Third, this project only considers one-step-ahead forecasts. It would also be useful to compare the models for longer forecast horizons. Finally, the feature importance values describe how strongly variables are used by the machine learning models and should not be interpreted as causal effects.

Overall, the random forest with lagged case values gives the lowest RMSFE. However, several linear models also perform competitively, and most differences are not statistically significant after the Bonferroni correction. The main practical result is that the weekly lag structure is very important, while adding more covariates or repeatedly updating the model does not always improve forecast accuracy.

References

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Breiman, L., Friedman, J. H., Olshen, R. A., and Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth, Belmont.
- Diebold, F. X. (2012). Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of diebold-mariano tests. NBER Working Paper 18391, National Bureau of Economic Research.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press, Princeton.
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with numpy. *Nature*, 585:357–362.
- Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.
- Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95.
- McKinney, W. (2010). Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61.
- Open-Meteo (2024). Historical Weather API. <https://archive-api.open-meteo.com>. Accessed July 2026.
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830.
- Petropoulos, F. and Makridakis, S. (2020). Forecasting the novel coronavirus covid-19. *PLOS ONE*, 15(3):e0231236.
- Robert Koch-Institut (2026). SARS-CoV-2 Infektionen in Deutschland. https://github.com/robert-koch-institut/SARS-CoV-2-Infektionen_in_Deutschland. Open-data repository; accessed July 2026.
- Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464.
- Seabold, S. and Perktold, J. (2010). statsmodels: Econometric and statistical modeling with python. In *Proceedings of the 9th Python in Science Conference*, pages 92–96.

- Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). Scipy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272.
- Waskom, M. L. (2021). seaborn: Statistical data visualization. *Journal of Open Source Software*, 6(60):3021.